Designing Accessible Systems

Dyslexia & UI Research



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The Team



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Introduction

This project explores how HCI principles can make digital tools accessible for people with dyslexia. Standard web apps use cramped layouts and poor typography that trigger visual fatigue. Through user interviews, we are gathering insights to guide our design choices. Our goal is to use spacious layouts, strong contrast, and clear visual cues to remove reading barriers.

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Needfinding Methodologies

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The Interviewee



- Principal of Belize Christian Academy
- Chosen and recruited because of his experience as a principal as well as a teacher.
- Interviewed at BCA by Alfred

Mr. Acosta

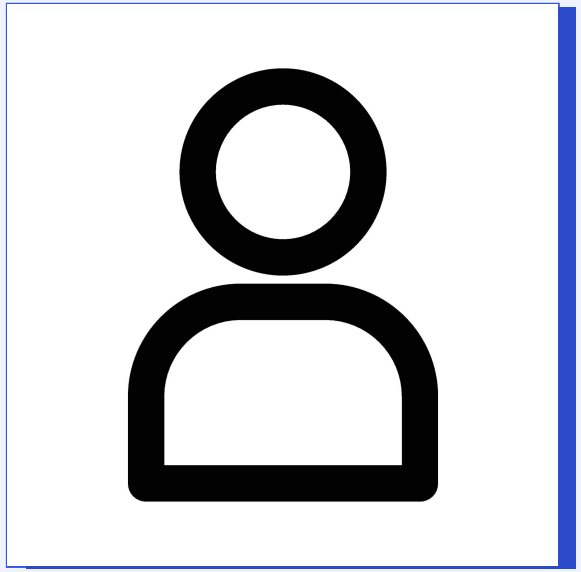


- Former teacher from Belmopan Comprehensive School
- Chosen and recruited because of her experience as a teacher.
- Interviewed at home by Javier.

Ms. Coyoc

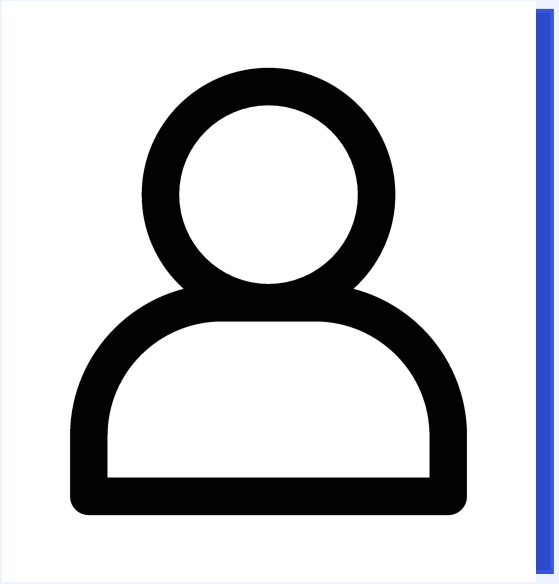
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The Interviewee



- Student of Belize Christian Academy
- Chosen and recruited because of their experience with dyslexia.
- Interviewed at BCA by Alfred

Student 1



- Student of Belize Christian Academy
- Chosen and recruited because of their experience with dyslexia.
- Interviewed at BCA by Alfred

Student 2

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Questions Asked (Teachers):

1. Describe an experience where a student with dyslexia struggled to navigate or understand a complex course gradebook or syllabus layout.
2. How different is it when you are teaching a student with dyslexia compared to others without dyslexia? What methods do you have to use to ensure they learn effectively?
3. Share a time when providing visual progress elements (like progress bars or clear category cards) noticeably helped dyslexic students track their coursework.



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Questions Asked (Students):

1. Describe an experience of how dyslexia impacted your learning or daily life, whether that be in the classroom or using a website.
2. What device (phone, laptop, tablet) do you usually use when checking course materials, and what issues have you faced using that device for school apps?
3. Describe a situation where a visual layout (such as low contrast text, distracting backgrounds, or inconsistent fonts) made a web page hard to read.
4. Describe a time when using a website with clear visual cues (like bold headings, colored badges, or distinct borders) helped you process information much faster.

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Interview Results

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1

Describe an experience where a student with dyslexia struggled to navigate or understand a complex course gradebook or syllabus layout.

My student with dyslexia struggled with our dense course layout, getting confused by scattered tracking and weighted grades. They lost track of completed work and syllabus details. To support them, I chunked content into smaller sections with clear headings, consistent formatting, and verbal explanations.

- Mr. Acosta

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2

How different is it when you are teaching a student with dyslexia compared to others without dyslexia? What methods do you have to use to ensure they learn effectively?

It's not that hard to teach a student that doesn't have dyslexia but when teacher teach students with dyslexia they have to go slower with the topics and have to listen and be patient with these types of students.

- Ms. Coyoc

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1 Describe an experience of how dyslexia impacted your learning or daily life, whether that be in the classroom or using a website.

Having dyslexia causes severe short-term memory issues and slows down my learning process. When doing online research, I almost always forget what I am searching for or why I started.

- Student 1

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2

Describe a time when using a website with clear visual cues (like bold headings, colored badges, or distinct borders) helped you process information much faster.

When I was in a rush checking a website, the bold headings significantly helped me locate where I was supposed to be.

- Student 2

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Analysis

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Empathy chart

Say

"I had a student with dyslexia who struggled to navigate a course work because there was a lot of information presented in small sections"

"when the teacher teaches students with dyslexia they have to go slower with the topics and have to listen and be patient"

"The student also had difficulty following a syllabus when important information was presented in long paragraphs"

"I noticed that students with dyslexia often lagged behind classmates"

Think

What's the best way to accommodate a student with dyslexia?

How can I ensure the student understands the material?

Recalling experiences with teaching dyslexic students and understanding their needs.

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Empathy chart

Do

I remember teaching a student with dyslexia and I had to use bright coded cards to help them with their course work.

A student progress was presented with visual elements motivated the student to continue thriving to succeed in her course work.

Broke the information into smaller sections, used clear headings and consistent formatting.

Asked student to verbalize what their assigned task was.

Feel

Wanting to help students learn effectively.

Feeling concerned for the education of students with dyslexia.

Unsure or unequipped on how to help students due lack of training & equipment.

Struggling with educating students with dyslexia.

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Insights and Needs

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"To support them, I broke the information into smaller sections, used clear headings and consistent formatting..."

- **Mr. Acosta**

Insight:

Complex layouts force teachers to manually simplify course content.

Need:

Automated systems that chunk data and highlight key tasks.

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"When teachers teach students with dyslexia, they have to go slower with the topics and have to listen and be patient."

- **Ms. Coyoc**

Insight:

Standard digital layouts present data too quickly and densely for dyslexic students to process smoothly.

Need:

Simplified, digestible interface elements that let students absorb information at their own pace.

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"Almost 99% of the time I be forgetting why or what am I researching for..."

- **Student 1**

Insight:

Dyslexia impairs short-term memory during multi-step tasks.

Need:

Clear navigation anchors and visible progress trackers.

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"The bold headings significantly helped me locate where I was supposed to be."

- **Student 2**

Insight:

Dense text causes rapid visual fatigue and reading confusion.

Need:

Spacious layouts with bold headings and visual cues.

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Questions About the Results

- How can digital tools make studying feel less exhausting for students with dyslexia?
- Is there a simple way to keep students from getting lost when switching between tasks?
- What interface elements give dyslexic students the most confidence while navigating school sites?

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Summary

- 4 interviews were conducted, two with teachers who have had experience teaching students with dyslexia and two with students who have dyslexia
- Learned about different methods that teachers used to teach students with dyslexia
- Gained an understanding on the website features that students with dyslexia like and dislike

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Thank you!

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